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Iterative CAD files Overview

Video Demonstration

Exploded View Video: https://youtu.be/3uYHuE0vSzA

Electronic Schematics

Multiple changes to the scope of the project were made for the electronic components. Initially, the components were selected to facilitate 5 unique games, with dedicated sensors and actuators for most. Since the project focused primarily on game design, its scope was initially quite large. However, as the final deadline approached, technical feasibility was assessed, and costs were factored in; two final sensors and actuators were chosen. Figure 1 illustrates the initial schematic used to construct the final prototype.

The first sensor included in the final prototype is the 3x3 matrix keypad. The keypad was chosen for numerous reasons. First, the keypad allows the concept of the controller to meet the users' expectations of a “bomb-defusal” game. As discussed in the final report, bombs in popular media often depict bombs with a keypad on the front. By using a keypad, the device could leverage established design conventions to communicate the purpose of the device more easily. Next, the keypad was favoured for the number of buttons on the keypad, which supports a wide variety of games.

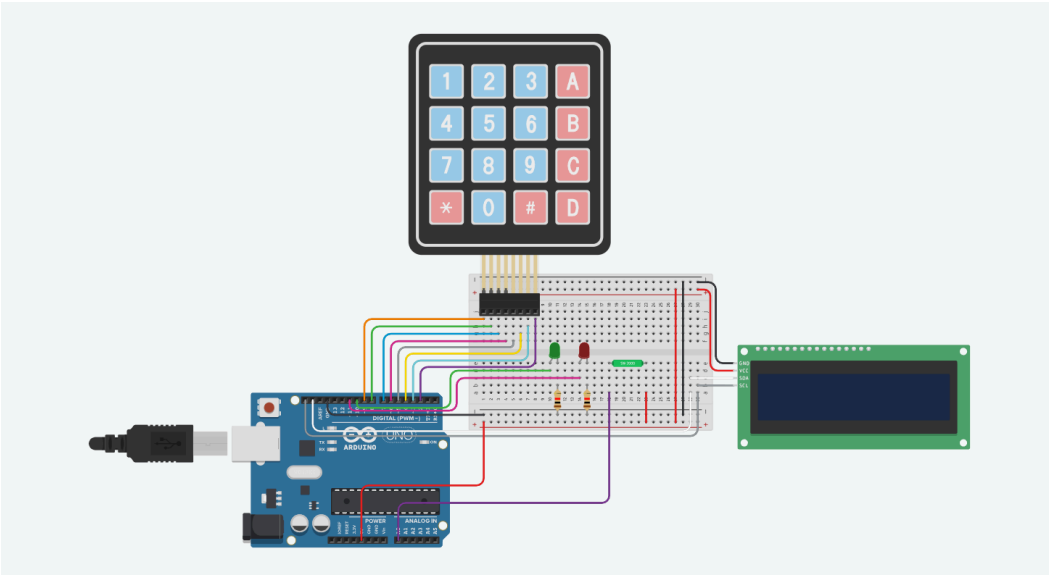
The second sensor included in the final prototype of this controller was an accelerometer. Because this project focuses on “unconventional” gameplay not commonly associated with bomb-defusal, an accelerometer was selected to surprise the user. The accelerometer is used in the “shake” game, where the user must shake the controller vigorously to increase a progress bar.

Next, the first actuator chosen for the controller was the I2C LCD. For the same reason mentioned for the matrix keypad, the LCD aids the controller in bridging the user's mental gap between bombs in media and this controller. The LCD display also facilitates multiple games, making it a versatile choice for this project.

Lastly, the second actuator chosen for this project is the speaker. The speaker allows the controller to play a variety of sounds conveying its state to the user. Sounds are played through the speaker from a micro SD bus, and range from conventional ominous bomb ticking sounds to more unfamiliar sounds. For reasons previously mentioned, combining both conventional bomb iconography with unconventional new elements unique to this controller creates a sense of surprise in the user.

Figure 1

An electronic schematic made in Tinkercad



Note. This schematic is missing an accelerometer and a speaker. These omissions were made because the scope of the project had not been finalized.

3D Assembly

Blueprints/Drawings

The following two drawings were made using Fusion 360's drafting tools. Custom housing for each component of this controller was designed and iterated on separately. Each iteration brought the housing closer to the required dimensions of each component. For example, iterations changed the size of screw holes.

Figure 2 is the blueprint for the Lid of the controller. On the lid is the housing for the LCD display and the Keyboard. Once both housing designs were finalized, the two components were attached to the Lid of the device.

Figure 2

A technical drawing of the Lid

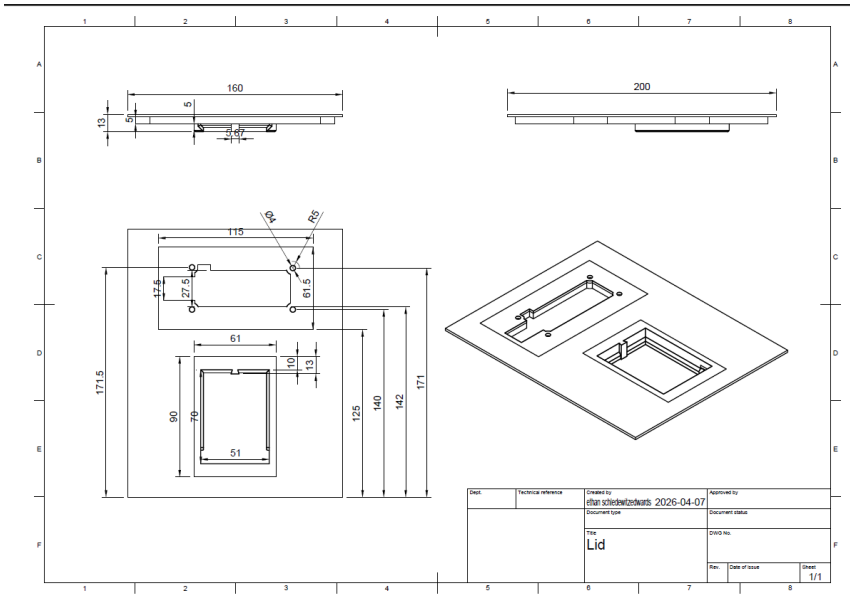
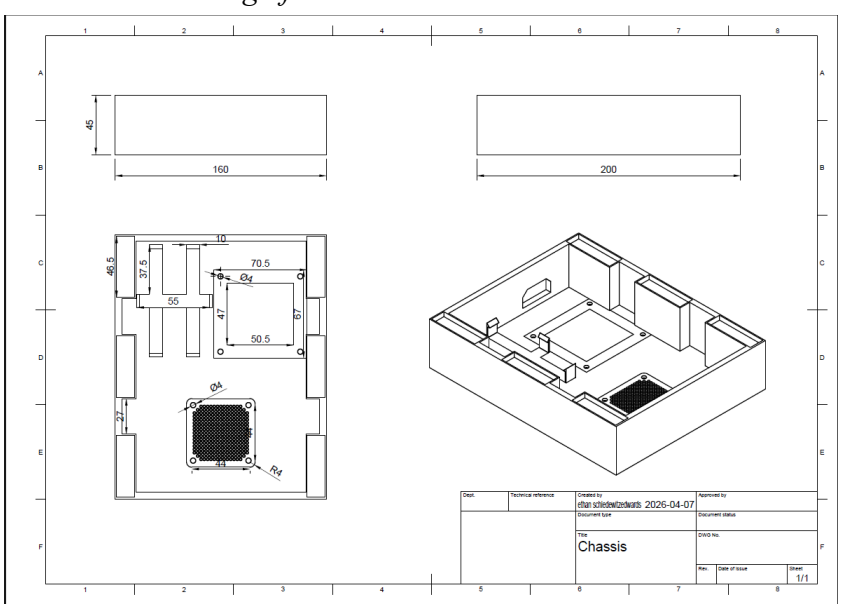


Figure 3 is the blueprint for the Chassis of the controller. The Chassis includes six magnets to fasten it to the bottom of the Lid. The chassis features a custom mount for the breadboard, ESP32, and speaker. The three custom mounts were iterated on separately and added to the Chassis once they were finalized.

Figure 3

A technical drawing of the Chassis

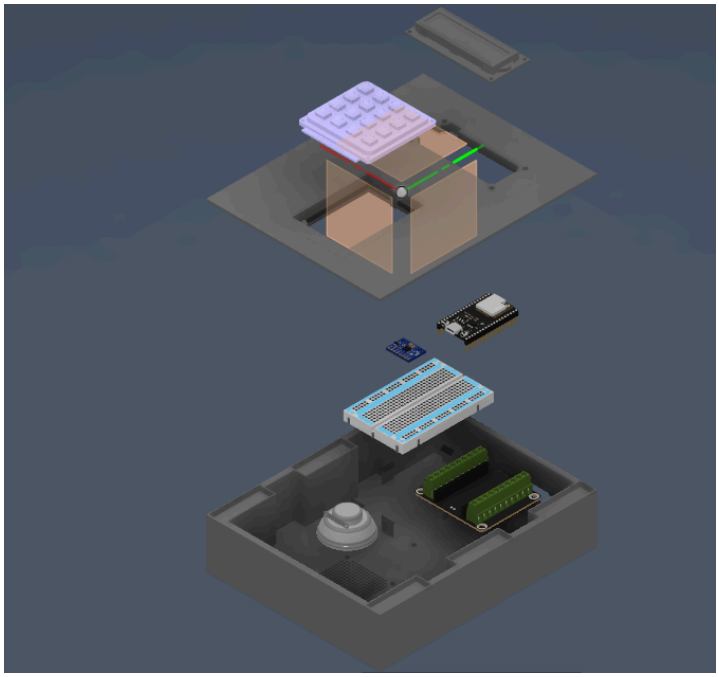


Exploded View

Figure 3 showcases the exploded view of our controller. The image was captured from our “Controller Exploded View” video, which was rendered using Fusion 360’s animation tools.

Figure 3

A screenshot of the Exploded View from the rendered video

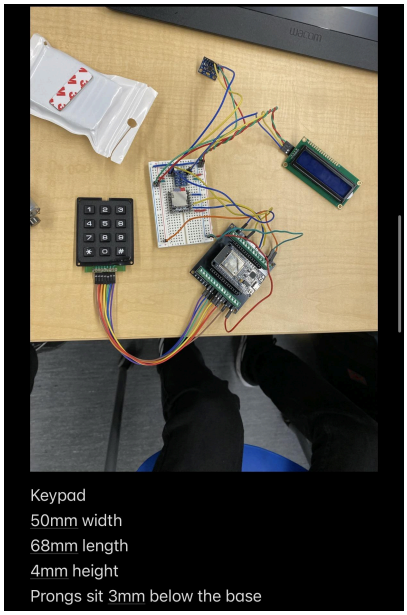


Approach on Iteration

To ensure that all the electronic components and wires fit neatly within our controller's exterior, careful measurements were taken for each part using calipers. Each measurement also detailed recommended clearance for wiring, ensuring the shell could contain absolutely everything. For example, Figure 4 illustrates the measurements taken for the 3x3 Matrix Keypad with a suggested clearance for wiring to run 3mm below the component.

Figure 4

Measurements for the keypad, taken using calipers



Using these measurements, individual housing for each component was prototyped using Fusion 360. For instance, Figure 5 showcases multiple iterations of housing designed specifically for the LCD screen. Figure 6 highlights multiple iterations for the Keypad and Speaker housing.

Figure 5

Iterations on the housing for the LCD screen

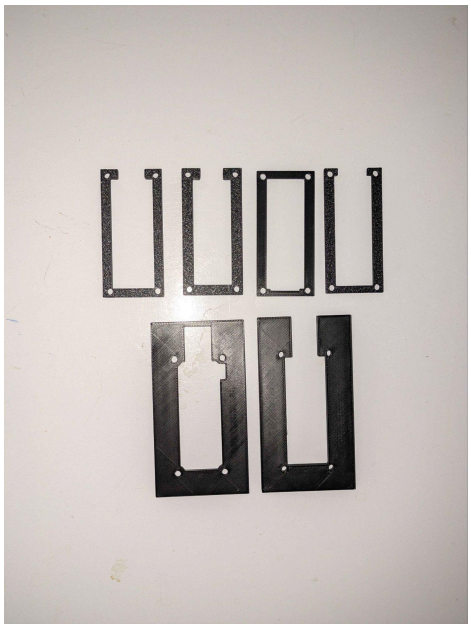
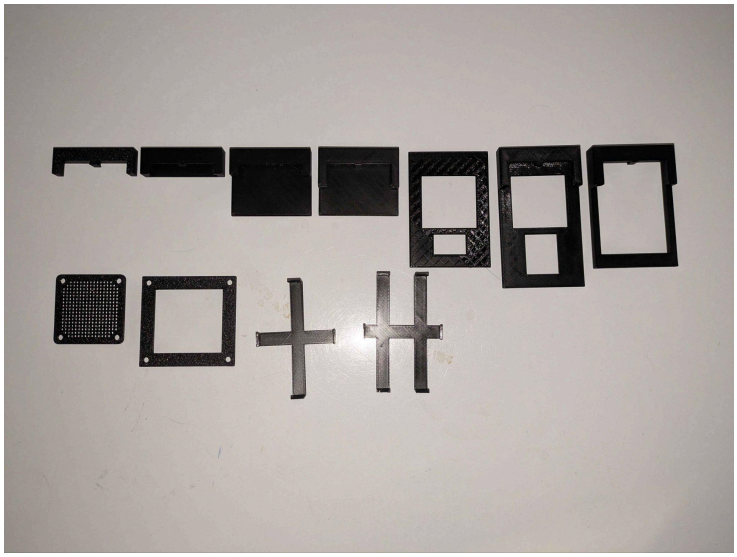


Figure 6

Iterations on the housing for the keypad, speaker, and breadboard



Note. The top row of prototypes is solely for the Keypad, the bottom row features two iterations for the speaker housing, and three iterations for the breadboard housing.

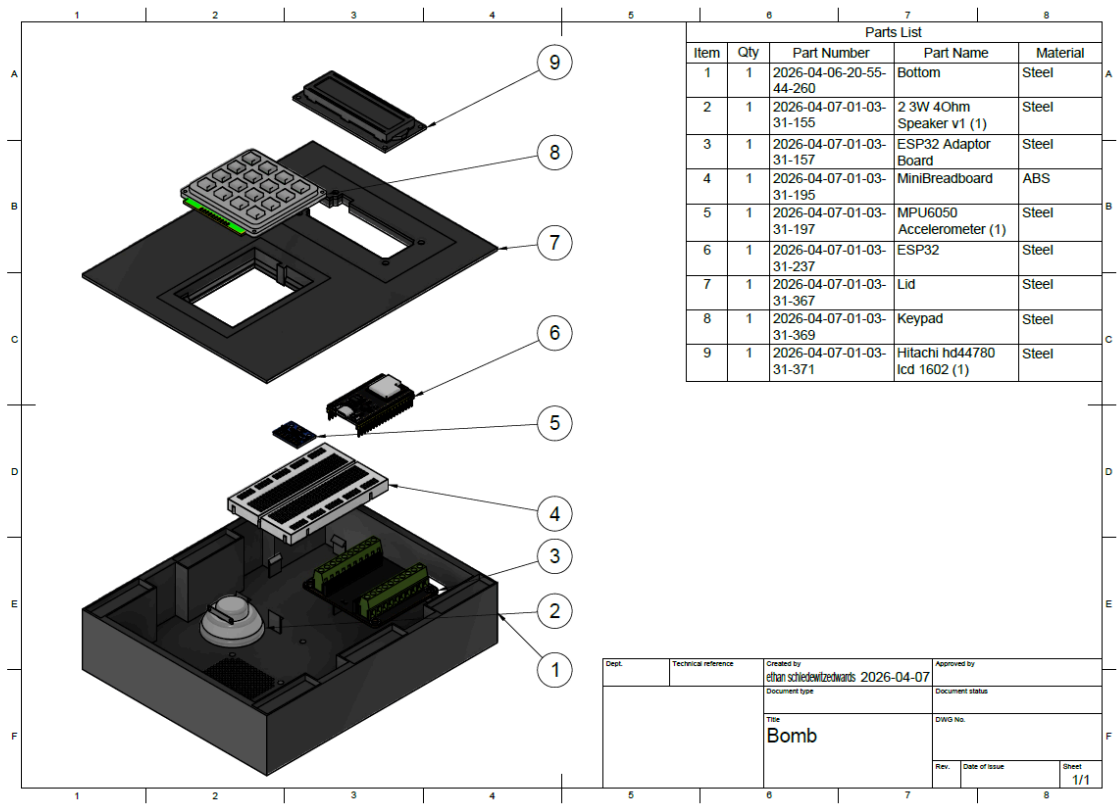
The design of the housing for each component was iterated on extensively, ensuring it fit each component optimally. Once each individual component had housing designed for it, all components were combined into the design for the final Chassis and Lid of this controller prototype. Much like the housing for the individual components, both the Chassis and Lid underwent multiple iterations to ensure everything fit nicely.

Bill of Materials

After the design of the Lid and Chassis of the controller was finalized, the Bill of Materials visualized in Figure 7 was created using Fusion 360's drawing tools.

Figure 7

The final Bill of Materials



Item	Qty	Part Number	Part Name	Material
1	1	2026-04-06-20-55-44-260	Bottom	Steel
2	1	2026-04-07-01-03-31-155	2 3W 4Ohm Speaker v1 (1)	Steel
3	1	2026-04-07-01-03-31-157	ESP32 Adaptor Board	Steel
4	1	2026-04-07-01-03-31-195	MiniBreadboard	ABS
5	1	2026-04-07-01-03-31-197	MPU6050 Accelerometer (1)	Steel
6	1	2026-04-07-01-03-31-237	ESP32	Steel
7	1	2026-04-07-01-03-31-367	Lid	Steel
8	1	2026-04-07-01-03-31-369	Keypad	Steel
9	1	2026-04-07-01-03-31-371	Hitachi hd44780 lcd 1602 (1)	Steel

Note. Some of the models used to represent the electronic components are incorrect and were used as approximations for the exploded view and bill of materials. The incorrect components are the matrix keypad and LCD screen.

References

[LCD]
https://grabcad.com/library/hitachi-hd44780-lcd-1602-1

[ESP32]
https://grabcad.com/library/esp32-devkit-32u-with-external-antenna-and-3-types-of-adapter-boards-1

[Breadboard]

https://grabcad.com/library/minibreadboard-2

[Accelerometer]
https://grabcad.com/library/mpu6050-accelerometer-module-2

[Speaker]
https://grabcad.com/library/3w-4ohm-speaker-1

[Keypad]
https://grabcad.com/library/4x4-button-keypad-matrix